INJECTIVES AND K-INJECTIVES RESTRICT FROM THE BIG ETALE SITE TO THE SMALL ONE

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-compare-injectives (source lines 799-820); the proof environment beginning on the line after the following blank line (source line 822) is its own proof and closes at source line 851.

Editorial note. The proof cites nine results, all outside the chapter (Homology, Derived Categories, Topologies twice, Modules on Sites three times, Cohomology on Sites twice, Etale Cohomology); each is declared as a tagged source pointer, and no in-chapter context is needed.

Editorial note. The author gives two independent proofs of the substantive parts and says so explicitly ('First proof', 'Second proof'); both are reproduced verbatim.

Editorial note. The retained preamble is the chapter's own (sources/stacks-spaces-more-cohomology/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup and theorem declarations; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's chapter Cohomology of Algebraic Spaces, counted once: the restriction of injectives and K-injectives from the big etale site of algebraic spaces over X to the small etale site of Y and of X.

Allowed standard prerequisites (editorial).

- Right adjoints of exact functors preserve injectives. If a functor is a right adjoint of an exact functor, it preserves injectives and K-injectives (Homology and Derived Categories, declared as tagged source pointers).. It gives parts (1)(b) and (2)(b) at source lines 823-828.
- The lower shriek adjunction and its exactness. The functor i_{f,!} is a left adjoint of i_f^{-1}, constructed for sheaves of sets and abelian sheaves, and it is exact (Topologies and Modules on Sites, declared as tagged source pointers).. It is the first proof of parts (1)(a) and (2)(a) at source lines 831-838.
- Factorization of i_f through the small site and localization. i_f = i_Y o f_{big}, and f_{big} is a localization so pullback by it preserves injectives and K-injectives (Topologies and Cohomology on Sites, declared as tagged source pointers).. It is the second proof of parts (1)(a) and (2)(a) at source lines 839-846.
- The failure for modules. The analogous statements for modules of O-modules fail, as shown by an explicit counterexample (Etale Cohomology,

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declared as a tagged source pointer).. It is the closing remark of the statement and proof at source lines 819 and 848-850.

Source reference. Original lemma 015Z: sources/stacks-chow/tags.txt, source lines 1526–1526. Cross-chapter pointer whose target carries the canonical tag 015Z in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 08BJ: sources/stacks-chow/tags.txt, source lines 10569–10569. Cross-chapter pointer whose target carries the canonical tag 08BJ in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 021F: sources/stacks-chow/tags.txt, source lines 2668–2668. Cross-chapter pointer whose target carries the canonical tag 021F in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04BG: sources/stacks-chow/tags.txt, source lines 5651–5651. Cross-chapter pointer whose target carries the canonical tag 04BG in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 04BH: sources/stacks-chow/tags.txt, source lines 5652–5652. Cross-chapter pointer whose target carries the canonical tag 04BH in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 021I: sources/stacks-chow/tags.txt, source lines 2671–2671. Cross-chapter pointer whose target carries the canonical tag 021I in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 03F3: sources/stacks-chow/tags.txt, source lines 4478–4478. Cross-chapter pointer whose target carries the canonical tag 03F3 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 08FI: sources/stacks-chow/tags.txt, source lines 10711–10711. Cross-chapter pointer whose target carries the canonical tag 08FI in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 0758: sources/stacks-chow/tags.txt, source lines 9122–9122. Cross-chapter pointer whose target carries the canonical tag 0758 in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Lemma 0.1. *Let S be a scheme. Let $Y \to X$ be a morphism of $(Spaces/S)_{étale}$.*

1. *If $\mathcal{I}$ is injective in $Ab((Spaces/X)_{étale})$, then*
   (a) *$i_f^{-1}\mathcal{I}$ is injective in $Ab(Y_{étale})$,*
   (b) *$\mathcal{I}|_{X_{étale}}$ is injective in $Ab(X_{étale})$,*
2. *If $\mathcal{I}^\bullet$ is a K-injective complex in $Ab((Spaces/X)_{étale})$, then*
   (a) *$i_f^{-1}\mathcal{I}^\bullet$ is a K-injective complex in $Ab(Y_{étale})$,*

(b) $\mathcal{I}^\bullet|_{X_{étale}}$ *is a K-injective complex in* $Ab(X_{étale})$,

The corresponding statements for modules do not hold.

Proof. Parts (1)(b) and (2)(b) follow formally from the fact that the restriction functor $\pi_{X,*} = i_X^{-1}$ is a right adjoint of the exact functor π_X^{-1}, see Homology, Lemma 015Z and Derived Categories, Lemma 08BJ.

Parts (1)(a) and (2)(a) can be seen in two ways. First proof: We can use that i_f^{-1} is a right adjoint of the exact functor $i_{f,!}$. This functor is constructed in Topologies, Lemma 021F for sheaves of sets and for abelian sheaves in Modules on Sites, Lemma 04BG. It is shown in Modules on Sites, Lemma 04BH that it is exact. Second proof. We can use that $i_f = i_Y \circ f_{big}$ as is shown in Topologies, Lemma 021I. Since f_{big} is a localization, we see that pullback by it preserves injectives and K-injectives, see Cohomology on Sites, Lemmas 03F3 and 08FI. Then we apply the already proved parts (1)(b) and (2)(b) to the functor i_Y^{-1} to conclude.

To see a counter example for the case of modules we refer to Étale Cohomology, Lemma 0758. □